

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 1 of 6

(1) Research Program/Project/Study Title: QUALITY CHARACTERISTICS AND SENSORY EVALUATION OF NOODLES INCORPORATED WITH DIFFERENT KINDS OF FERMENTED SHELLFISH	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Pontevedra	
Province:		Capiz	
Region:		6	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food Sector		(8) Discipline/Field of Specialization Food Science and Technology	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 2 of 6

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Noodle products like bihon, sotanghon, pasta, canton, fresh miki and rice noodles or miswa are popular food commodities among Filipinos (PSA, 2016). In fact, instant noodles is a favourite and our country was currently reported to be at the top seventh among the world consumers' (Wilkinson et al., 2019). The global acceptance of noodles could be attributed to its affordable costs, palatability, stability, and convenience according to Ikotun (2013). Adejuwon, Jideani, & Falade, (2020) however imparted that despite its wide acceptance there are disputable perceptions concerning its effect on people's health. It was further revealed in their paper that research efforts on noodles are directed towards fortification with protein, minerals and vitamins to improve its nutritional value. With its rising popularity, noodles then can be an appropriate food vehicle for potential fortificants of local origin which could facilitate enhancement of the product in terms of its nutritive aspect.

On the otherhand, fermented shellfishes are a good source of protein, amino acids, and polypeptides that are important to health (Martínez-álvarez et al., 2017). This can be utilized as a fortificant which could serve both as a salt replacement at the same time flavor enhancer to noodles. The technology of fortifying noodles with fermented shellfishes could lead to an improved food product, which is both deliciously satisfying and at the same time meet the nutritional need of the community. This research endeavour could help address the need for an accessible nutritious food and is anchored on sustainable development goal (SDG) 3: Good Health and Well-being. This will not only help the noodles industry but it will also promote the utilization and therefore the marketability of fermented shellfish which are abundant in our region.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

Experimental study design, particularly Randomized Complete Block Design (RCBD) will be employed in the conduct of the study. Noodles will be prepared using (5) experimental treatments and the experiments will be done in three trials. The formulation of the noodles using different fermented shellfishes will serve as the experimental variable, while drying temperature will serve as the blocking. The noodles produced in the experimentation stage will be subjected to the different assays which will become the basis of evaluation of the product's quality and characteristics. Test will be performed using standard procedures. Statistical analysis and interpretation of results on the gathered data will be conducted to answer the objectives of the study.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

In general, the proposed study aims to assess the quality, sensory property and yield of the noodles incorporated with fermented shellfish.

Specifically, it aims to:

1. describe the microscopic structure of the noodles;
2. determine the percent moisture content (%MC) and percent water uptake on rehydration of the noodles;
3. evaluate the chemical composition of the noodles in terms of its proximate composition and histamine content;
4. evaluate the sensory characteristics of the noodles;
5. evaluate if significant difference exist among noodles incorporated with fermented shellfish in terms of %MC, % water uptake, proximate composition, and sensory characteristics;

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 3 of 6

6. conduct material balance and yield analysis of the prepared noodles;
7. conduct shelf life study of the noodles incorporated with fermented shellfishes; and
8. conduct consumer's acceptability test

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

A number of studies has been done on food fortification. In the review paper of Udachan et al. (2018), noodles and other wheat flour-based foods such as breads and biscuits have been fortified with protein and fiber to enhance its nutritional and functional properties using plant proteins and fish proteins. Meanwhile, by-products of fruit processing, becomes the source of fiber source used in the fortification. The fortification results to several changes on the chemical composition as well as the physical properties of the food product.

Gallegos-Infante et al., (2010) found out that adding common bean flour in spaghetti increases the protein and resistant starch contents as well the indigestible insoluble fraction of the product. However it lowers the total starch, and the available starch in the pasta. Also it was observed that the addition of common bean flour lowers cooking time but increases cooking loss. These changes in the spaghetti has positive implications for human health.

In the study of Litaay et al. (2022), non-wheat noodles made of sago flour was fortified with anchovy flour resulting to an increase on its nutritional content. In another study soft wheat flour noodles was incorporated with cucumber pomace which improve the cooking quality, the nutritional value, sensorial capacity and textural characteristics of the noodles(Saad et al., 2021).

Several patents were found on fortification of noodles, both as product and method. It includes the iron-fortified fine dried noodles and its method, dietary fiber-fortified noodles and its method, spinach noodles added with celery(WIPO - Search International and National Patent Collections, n.d.)

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

Modification in the formulation of a food will result to a change on the product's quality. Quality attributes that may be altered in the course of varying either its ingredients or its method may include the following:physical characteristics, organoleptic attributes, chemical composition, microbiological and nutritional properties. This study will investigate the type of fermented shellfish that will be incorporated in the noodles as a salt replacement which is suppose to have an effect on the physico-chemical properties, chemical and sensory characteristics of the noodles.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Variables of the study:

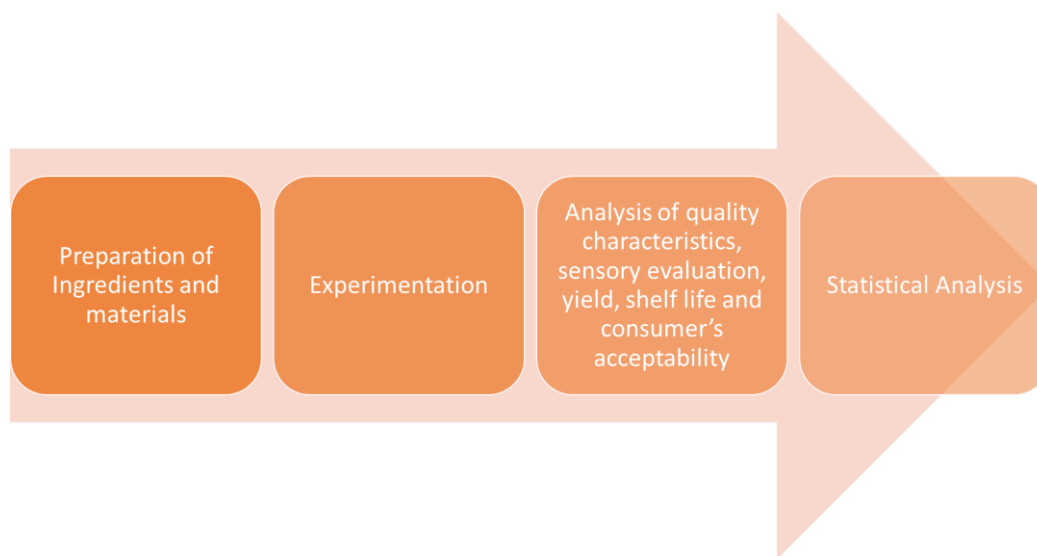
Independent – type of fermented shellfish & drying temperature

Dependent variables – microscopic structure, physico-chemical characteristics, chemical composition and sensory attributes.

Experimental Design: Randomized Complete Block Design (RCBD)

Treatments: A - control (standard formulation), flour, water and salt
B – fermented oyster
C – fermented bahong
D – flour, egg, fermented oyster
E – flour, egg, fermented bahong
F – commercial noodles

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 4 of 6



(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Activity	July				August				September			
	1	2	3	4	1	2	3	4	1	2	3	4
Refinement of proposal	■	■										
Purchase and preparation of materials			■	■								
Experimentation					■	■	■	■				
Data gathering					■	■	■	■				
Statistical analysis and interpretation								■	■	■		
Writing of proposal											■	■

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

1. Noodles fortified with fermented shellfishes
2. Standard formulation and method of preparing noodles with fermented shellfish
3. Utility model/patent of the process and product
4. Information and educational materials for the production of low-sugar beverages
5. Article/paper for publication
5. Process technology for extension activities

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Training on the processing methods of noodles making

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Farmers, food processors, health enthusiasts, consumers, entrepreneurs, researchers

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 5 of 6

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Leah B. Yakut	Project leader	Food Science and Technology
Reeven Simon	Member	Food service management
John Keneth Ada	Member	Business Administration
Mark Borines	Research Assistant	Food Technology

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 15 hrs/wk

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1@11,000/mo for 3 months	33,000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		5,000
- Calamansi		1,000
- Sugar		1,500
- Water		500
- 250 mL plastic bottle (PET)		3,000
- utility box, food containers, aluminum foil, sensory cups, sticker paper, dishwashing liquid, sanitizing agent, alcohol and other laboratory supplies		5,000
- Water activity meter		20,000
- Chromameter/Colorimeter		15,000
- plate count agar		2,500
- yeast and mold count agar		2,500
- reagents (NaOH, Na ₂ S ₂ O ₃)		5,000
- bond paper (6 reams)		1,500
- Miscellaneous (for incidental expenses)		5,000
- other office supplies (ink, ballpens, pencils)		5,000
C. CAPITAL OUTLAY		
Total		105,500.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Adejuwon, O. H., Jideani, A. I. O., & Falade, K. O. (2020). Quality and Public Health Concerns of Instant Noodles as Influenced by Raw Materials and Processing Technology. *Food Reviews International*, 36(3), 276–317. <https://doi.org/10.1080/87559129.2019.1642348>

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Gallegos-Infante, J. A., Bello-Perez, L. A., Rocha-Guzman, N. E., Gonzalez-Laredo, R. F., & Avila-Ontiveros, M. (2010). Effect of the addition of common bean (*Phaseolus vulgaris* L.) flour on the in vitro digestibility of starch and undigestible carbohydrates in Spaghetti. *Journal of Food Science*, 75(5).

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			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 6 of 6

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Litaay, C., Indriati, A., Sriharti, Mayasti, N. K. I., Tribowo, R. I., Andriana, Y., & Andriansyah, R. C. E. (2022). Physical, chemical, and sensory quality of noodles fortification with anchovy (*Stolephorus* sp.) flour. *Food Science and Technology (Brazil)*, 42, 1–7. <https://doi.org/10.1590/fst.75421>

Martínez-álvarez, O., López-Caballero, M. E., Gómez-Guillén, M. C., & Montero, P. (2017). Fermented Seafood Products and Health. *Fermented Foods in Health and Disease Prevention, January*, 177–202. <https://doi.org/10.1016/B978-0-12-802309-9.00009-1>

Saad, A. M., El-Saadony, M. T., Mohamed, A. S., Ahmed, A. I., & Sitohy, M. Z. (2021). Impact of cucumber pomace fortification on the nutritional, sensorial and technological quality of soft wheat flour-based noodles. *International Journal of Food Science and Technology*, 56(7), 3255–3268. <https://doi.org/10.1111/IJFS.14970>

Udachan, I. S., Pakhare, K. N., & Dagadkhair, A. C. (2018). Enhancement of Nutritional and Functional Characteristics of Noodles by Fortification with Protein and Fiber: A Review. *Journal of Pharmacognosy and Phytochemistry*, 7(1), 351–357.

Wilkinson, I., White, P., Carter, C., Kingwell, R., & Cowman, S. (2019). Wheat and barley markets in the Philippines - opportunities for Australia. *Australian Export Grains Innovation Centre, September*. <https://doi.org/10.13140/RG.2.2.13741.44002>

WIPO - Search International and National Patent Collections. (n.d.). Retrieved August 5, 2022, from https://patentscope.wipo.int/search/en/result.jsf?_vid=P21-L6GCGB-13791

(20) Capsule Curriculum Vitae (One-page brief C.V. only)